

SQL Server Beginner Class Guide

A simple training manual for junior developers

What is SQL?

SQL (Structured Query Language) is the language used to communicate with relational databases. It allows us to create, read, update, and delete data stored in tables.

What is SQL Server?

SQL Server is Microsoft's Relational Database Management System (RDBMS). It stores and manages data securely and efficiently.

Install SQL Server

1. Download SQL Server Developer Edition.
2. Run the installer.
3. Choose Basic Installation.
4. Complete setup and note the server name.

Install SQL Server Management Studio (SSMS)

1. Download SSMS.
2. Run the installer.
3. Open SSMS.
4. Connect to your SQL Server instance.

Database and Schema

Database stores data. Schema is a logical container inside a database.
Example: OfficeDB -> dbo -> Employee

Create Database

```
USE OfficeDB;
```

```
CREATE DATABASE OfficeDB;
```

Select Database

```
USE OfficeDB;
```

Create Table

```
USE OfficeDB;
```

```
CREATE TABLE dbo.Employee  
(  
  EmployeeId INT PRIMARY KEY,  
  EmployeeName VARCHAR(100),  
  Salary DECIMAL(10,2)  
);
```

Insert Data

```
USE OfficeDB;
```

```
INSERT INTO dbo.Employee  
(EmployeeId, EmployeeName, Salary)  
VALUES  
(1, 'John', 50000);
```

Select Data

```
USE OfficeDB;
```

```
SELECT *  
FROM dbo.Employee;
```

Select Specific Columns

```
USE OfficeDB;
```

```
SELECT EmployeeId, EmployeeName  
FROM dbo.Employee;
```

Where Clause

```
USE OfficeDB;
```

```
SELECT *  
FROM dbo.Employee  
WHERE Salary > 50000;
```

Order By

```
USE OfficeDB;
```

```
SELECT *  
FROM dbo.Employee  
ORDER BY Salary DESC;
```

Top Records

```
USE OfficeDB;
```

```
SELECT TOP 10 *  
FROM dbo.Employee;
```

Update Data

```
USE OfficeDB;
```

```
UPDATE dbo.Employee  
SET Salary = 55000  
WHERE EmployeeId = 1;
```

Delete Data

```
USE OfficeDB;
```

```
DELETE FROM dbo.Employee  
WHERE EmployeeId = 1;
```

Aggregate Functions

```
USE OfficeDB;
```

```
SELECT COUNT(*) AS TotalEmployees  
FROM dbo.Employee;
```

```
SELECT AVG(Salary) AS AverageSalary  
FROM dbo.Employee;
```

Group By

```
USE OfficeDB;
```

```
SELECT DepartmentId, COUNT(*)  
FROM dbo.Employee  
GROUP BY DepartmentId;
```

Inner Join

```
USE OfficeDB;
```

```
SELECT e.EmployeeName,  
d.DepartmentName  
FROM dbo.Employee e  
INNER JOIN dbo.Department d  
ON e.DepartmentId = d.DepartmentId;
```

Left Join

```
USE OfficeDB;
```

```
SELECT e.EmployeeName,  
d.DepartmentName  
FROM dbo.Employee e  
LEFT JOIN dbo.Department d  
ON e.DepartmentId = d.DepartmentId;
```

Subquery

```
USE OfficeDB;
```

```
SELECT *  
FROM dbo.Employee  
WHERE Salary >  
(  
SELECT AVG(Salary)  
FROM dbo.Employee  
);
```

View

```
USE OfficeDB;
```

```
CREATE VIEW dbo.vwEmployee  
AS  
SELECT *  
FROM dbo.Employee;
```

Index

```
USE OfficeDB;
```

```
CREATE INDEX IX_EmployeeName  
ON dbo.Employee(EmployeeName);
```

Stored Procedure

```
USE OfficeDB;
```

```
CREATE PROCEDURE dbo.GetEmployees  
AS  
BEGIN  
SELECT *  
FROM dbo.Employee;  
END;
```

Function

```
USE OfficeDB;
```

```
CREATE FUNCTION dbo.GetAnnualSalary  
(  
    @Salary DECIMAL(10,2)  
)  
RETURNS DECIMAL(10,2)  
AS  
BEGIN  
    RETURN @Salary * 12;  
END;
```

Trigger

```
USE OfficeDB;
```

```
CREATE TRIGGER dbo.trgEmployeeInsert  
ON dbo.Employee  
AFTER INSERT  
AS  
BEGIN  
    PRINT 'Employee Added';  
END;
```

Transaction

```
USE OfficeDB;
```

```
BEGIN TRANSACTION;
```

```
UPDATE dbo.Employee  
SET Salary = Salary + 1000  
WHERE EmployeeId = 1;
```

```
COMMIT;
```

Interview Questions

1. What is SQL?
2. What is SQL Server?
3. What is a Primary Key?
4. Difference between DELETE and TRUNCATE?
5. What is a JOIN?
6. What is a Stored Procedure?
7. What is an Index?